

Vinh Ly

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Summary

Software Engineer with **18 years** of experience building large-scale distributed systems at Google, YouTube, Whatnot, and Microsoft. Vinh Ly led backend infrastructure development at YouTube and Whatnot, designing real-time marketplace systems and video delivery infrastructure serving billions of users. At Google for over a decade, he managed multiple engineering teams that designed and maintained distributed systems powering Search, Ads, and Content Delivery. He brings end-to-end ownership of production systems from kernel-level optimization to end-user features. His next focus is AI platform development with **Python** and **Go**. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Work Experience

Staff Software Engineer | YouTube

02/2025 - Present | Remote

- Leading backend infrastructure development for YouTube core platform services, focusing on content delivery, real-time features, and API services at billion-user scale.

Staff Software Engineer | Whatnot

05/2024 - 02/2025 | Remote

- Designed real-time marketplace infrastructure handling concurrent transactions, live video streaming integration, and websocket-based notification delivery for 100K+ daily active users.
- Established data observability and monitoring pipelines for marketplace health metrics, reducing incident detection time by 40 percent through automated alerting.
- Built auction and bidding services with sub-second latency requirements, processing seven-figure daily transaction volume using Go microservices on Kubernetes.

Staff Software Engineer/Manager | Google

03/2015 - 04/2024 | Remote

- Designed and shipped distributed systems powering Google Search, processing 100 billion plus queries monthly with strict availability and latency guarantees for ranking and indexing pipelines.
- Optimized distributed caching and storage systems that cut infrastructure costs by 30 percent while improving p99 latency for core serving paths across search infrastructure.
- Directed engineering teams across Search, Ads, and Content Delivery infrastructure, managing 5 to 15 engineers across multiple product areas with end-to-end technical ownership.
- Led cross-functional initiatives to migrate legacy monolithic services to containerized microservices architecture, reducing deployment cycle time from weeks to daily releases.
- Partnered with product management and customer engineering teams to define technical roadmaps and translate product requirements into scalable system designs.

Software Engineer | Google

02/2011 - 02/2015 | Remote

- Built backend services for Google Search serving infrastructure using C++ and Python, designing distributed data pipelines processing billions of documents for indexing operations.

- Optimized performance-critical code paths in C++ for query processing, achieving measurable improvements in search latency and throughput for ranking algorithms.
- Contributed to the design of sharding and replication strategies for large-scale document storage systems supporting search ranking and retrieval operations.
- Developed infrastructure automation tooling for service deployment and configuration management, reducing operational overhead for distributed search serving clusters.

Software Design Engineer II | Microsoft

10/2007 - 12/2011 | Remote

- Developed distributed backend systems for enterprise product lines using C++, .NET, and SQL Server, building service-oriented architecture supporting high availability requirements.
- Designed and implemented data processing pipelines handling millions of daily transactions with fault tolerance and disaster recovery capabilities for production environments.
- Performed kernel-level debugging and performance analysis on Windows server infrastructure, identifying bottlenecks in I/O and memory subsystems affecting application throughput.

Software Design Engineer Intern | Microsoft

06/2006 - 08/2006 | Remote

- Contributed to distributed systems projects within the engineering organization, gaining hands-on experience with enterprise software development lifecycle and practices.

Education

B.S. Honors degrees in Computer Science

Virginia Tech | 2004 - 2007

CoreSkills

LanguagesandRuntimes: Python, Go, C++, Java, JavaScript, SQL, Bash, TypeScript, HTML, CSS

FrameworksandPlatforms: FastAPI, Django, gRPC, React, Node.js, Spring Boot

Infrastructure and Distributed Systems: Kubernetes, Docker, Distributed Systems, Microservices, Linux, Multi-Tenant Architecture

Data and Storage: PostgreSQL, MySQL, Redis, Kafka, Object Storage, Caching Systems, Data Pipelines

Observability and Reliability: Monitoring, Distributed Tracing, Alerting Systems, on-call, Terraform, AWS